

Xcode Essencial:
A Arte de Criar Apps Apple

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Propósito

Apresentar como funciona o processo de desenvolvimento de aplicativos para o ecossistema Apple, incluindo as etapas de criação, testes e publicação na App Store, além de explicar como se tornar um Apple Developer Member e quais benefícios e requisitos estão envolvidos nesse processo.

Whoami ?

Whoami ?

- Profissional +15 anos de experiência em Cyber
- Pesquisador em segurança nos ecossistemas: Windows Linux e Apple
- Apaixonado por Análise e Desenvolvimento de Malware
- Ajudo organizar Apple Red Team Village
- Fã de Chaves e Chapolin ☺





Topics

- Introdução ao Ecossistema Apple e ao Xcode
- Construindo Apps
- Apple Developer Program
- Simulação e Testes no Xcode
- Conclusão



Introdução ao Ecossistema Apple

Apple Ecosystem

macOS

Desktops
- Mac Pro
- Mac Mini
- iMac
- Mac Studio

Laptops
- MacBook Pro
- MacBook Air
- MacBook Neo

iOS

- iPhone
- iPod Touch

iPadOS

- iPad
- Mini
- Air
- Pro

watchOS

- Apple Watch

tvOS

- Apple TV (Gen 4)
- Apple TV 4k
- Apple TV HD

visionOS



Vision Pro
- Spatial Computing
- Immersive Experiences

Ecossistema Apple

- Linguagens principais:
  - Swift
  - Objective-C
- Frameworks Modernos:
  - SwitfUI
  - Ukit/AppKit

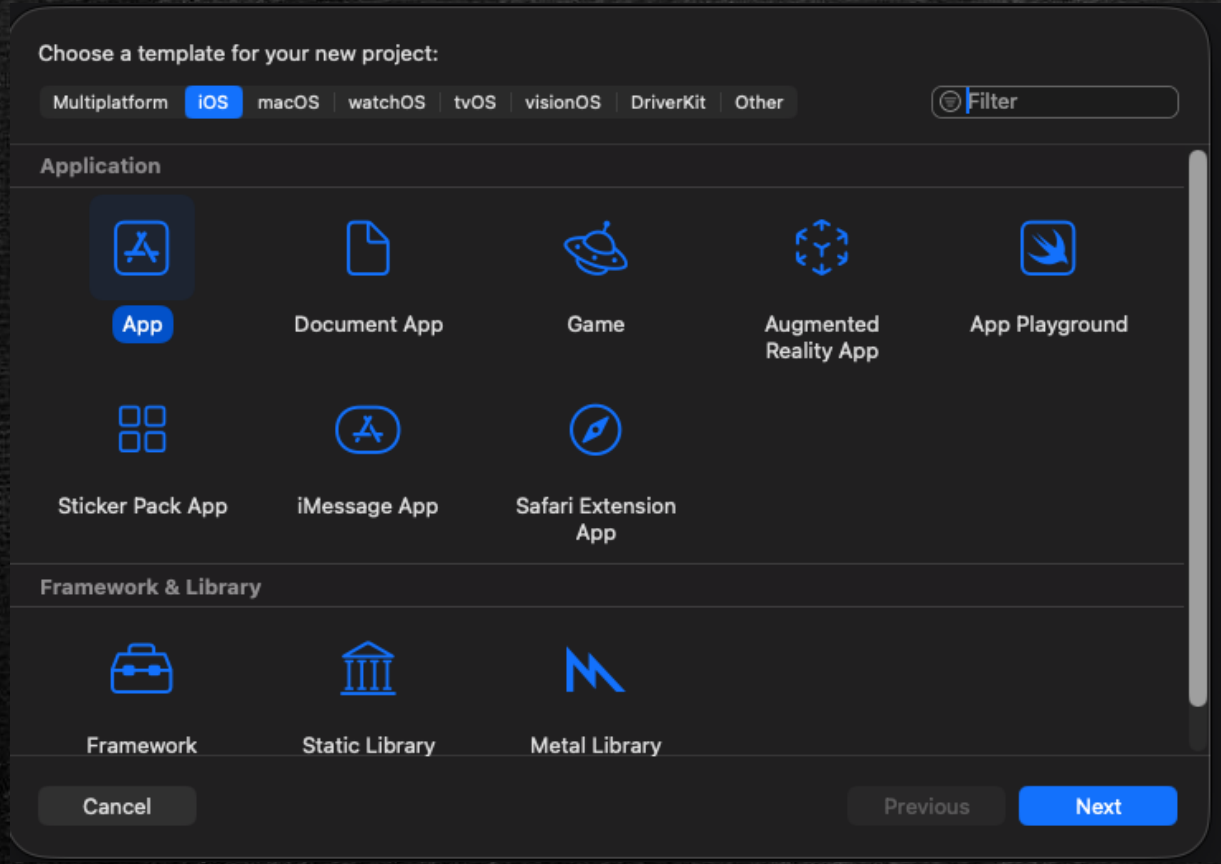
O que é o Xcode

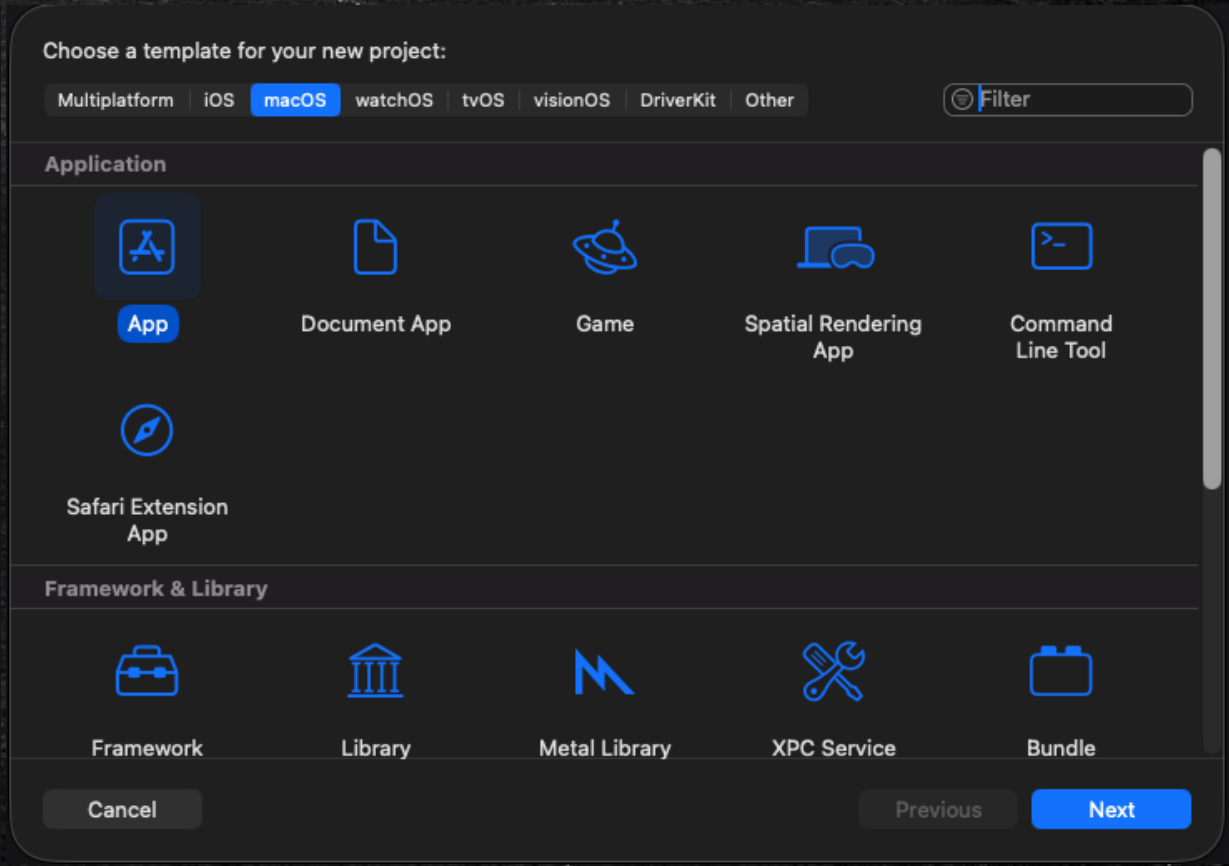
- IDE oficial da Apple:
  - Compilar Apps Apple
  - Gerenciar Simuladores
  - Assinar Apps
  - Integrar com App Store
  - Fazer Profiling e Debugging nativo
- Desenvolvimento Completo
- Design + Codigo + Testes + Publicação

O que é o Xcode

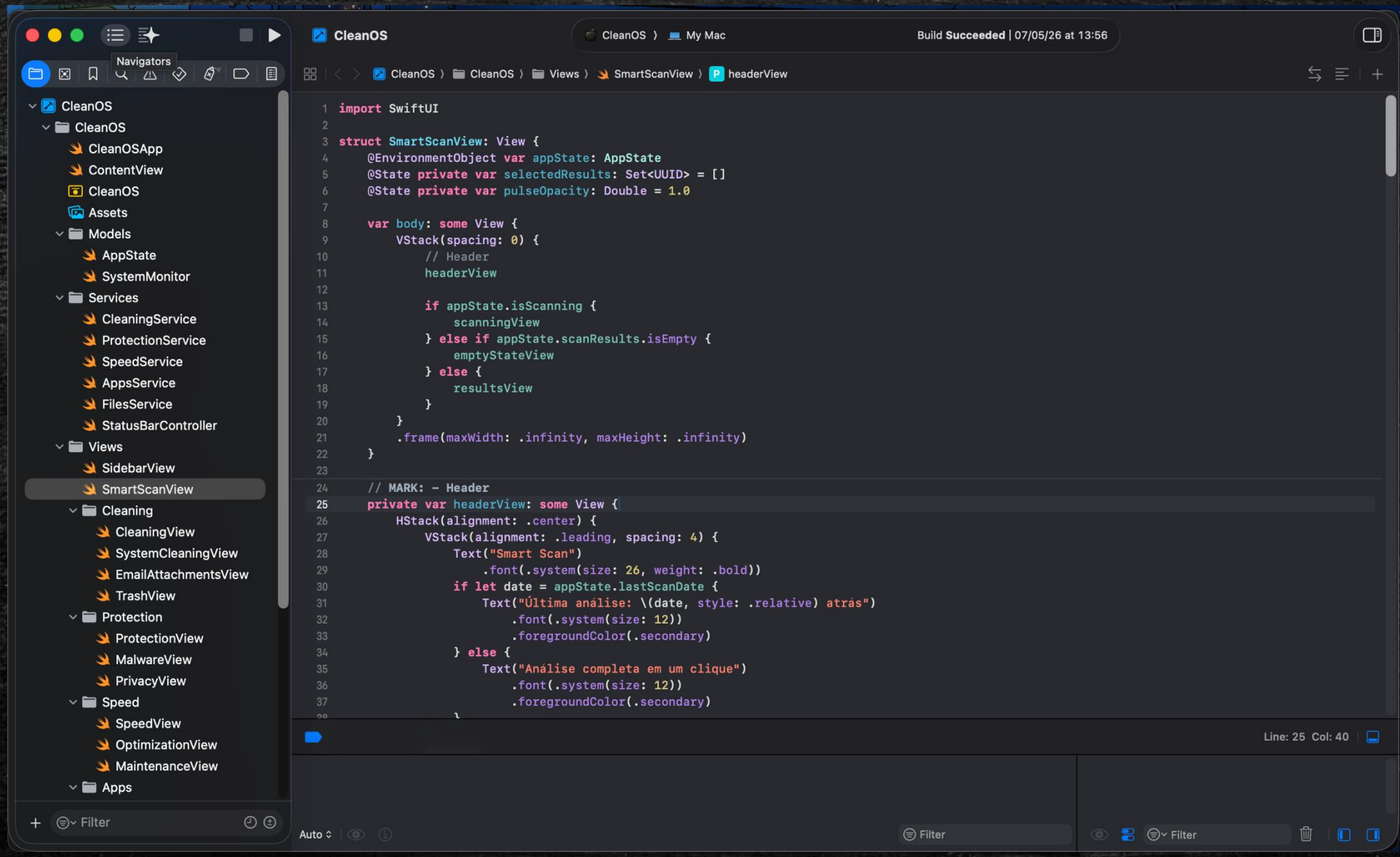


O que é o Xcode





O que é o Xcode



O que é o Xcode

- App (Aplicativo)
- SwitfUI (CleanOS)
- Scene (WindowGroup)
- View (Content: View)

```swift
import SwiftUI

struct ContentView: View {
    @EnvironmentObject var appState: AppState

    var body: some View {
        NavigationSplitView {
            SidebarView()
                .frame(minWidth: 220, idealWidth: 240, maxWidth: 260)
        } detail: {
            detailView
                .frame(maxWidth: .infinity, maxHeight: .infinity)
        }
        .navigationSplitViewStyle(.balanced)
        .background(Color(NSColor.windowBackgroundColor))
        .alert(appState.alertTitle, isPresented: $appState.showAlert) {
            Button("OK", role: .cancel) {}
        } message: {
            Text(appState.alertMessage)
        }
    }
```

```swift
import SwiftUI

@main
struct CleanOSApp: App {
    @NSApplicationDelegateAdaptor(AppDelegate.self) var appDelegate
    @StateObject private var appState = AppState()

    var body: some Scene {
        WindowGroup {
            ContentView()
                .environmentObject(appState)
                .frame(minWidth: 900, minHeight: 620)
        }
        .windowStyle(.hiddenTitleBar)
        .commands {
            CommandGroup(replacing: .newItem) {}
        }
    }
}
```



Construindo Apps

Apps - Xcode

- Bundle Identifier Único
  - *com.suaempresa.seuapp*
  - *Team Selecionado*
- Signig & Capabilities
  - *Criar Certificado*
  - *Gerar provisioning profiles*
  - *Assinar App*
- Version (1.0) e Build (1)
- App Icon
- Nome do App
- Privacy Usage Descriptions no Info.plist

Xcode Deploy

Status log for "CleanOS":

Processo de Distribuição Direta

Prepared archive for uploading	5 May 2026 at 22:07
Uploaded to Apple	5 May 2026 at 22:07
Ready to distribute	5 May 2026 at 22:08
Notarized CleanOS.app	
Notarized CleanOS.app	
Notarized CleanOS.app/Contents/MacOS/CleanOS	
Notarized CleanOS.app/Contents/MacOS/CleanOS	

Status log for "CleanOS":

Processo da App Store

Prepared archive for uploading	4 May 2026 at 15:04
Uploaded to Apple	4 May 2026 at 15:05

Xcode com Segurança

- Assinatura de Codigo
- Notorization
- SandBox
- Gatekeeper
- Entitlements (Permissões especiais no App)
- Hardened Runtime (Code Injection)
- App Privacy Details (Coleta de Dados)
- Xprotect

Apple Developer Program

Desenvolva. Inove. Publique.



ADP – Pre-requisitos

- Conta iCloud
- Cadastro no ADP (Apple Developer Program)
- Taxa de US$99,00 (Dolares)
- Envio de Documentação (CNH ou Passaporte)
- Confirmação da Documentação
- Após 48hs acesso ao programa é liberado

Apple Developer Program

Program resources

App Store Connect

Manage your app's builds, metadata, and more on the App Store.

- Apps
- App Analytics
- Sales and Trends
- Payments and Financial Reports
- Business
- Users and Access

Certificates, IDs & Profiles

Manage the certificates, identifiers, profiles, and devices required to develop, test, and distribute apps.

- Certificates
- Identifiers
- Devices
- Profiles
- Keys
- Service configuration

Services

View and manage your usage of developer services.

- Xcode Cloud
- CloudKit
- Apple Maps
- Push Notifications
- WeatherKit

Apple Developer Program

- Distribuição de apps na Apple Store
- Distribuição Privada/Empresarial
- Acesso a versões Betas e Ferramentas exclusivas
  - iOs, macOS, Xcodes e APIs Betas
- TestFlight (Teste Betas)
- Analytics e Métricas da App Store
- Monetização Integrada
- Publicação em Multiplas Plataformas

App - Não Assinado

```
zoziel:Downloads punisher$ codesign -dv --verbose=4 MacSafe.app
Executable=/Users/punisher/Downloads/MacSafe.app/Contents/MacOS/MacSafe
Identifier=MacSafe
Format=app bundle with Mach-O thin (arm64)
CodeDirectory v=20400 size=80288 flags=0x2(adhoc) hashes=2503+3 location=embedded
VersionPlatform=1
VersionMin=720896
VersionSDK=786688
Hash type=sha256 size=32
CandidateCDHash sha256=6611aefb3b6f5a6711a5f8bec81dc88a7ed8108c
CandidateCDHashFull sha256=6611aefb3b6f5a6711a5f8bec81dc88a7ed8108cb86aa972d2b23320fed169d1
Hash choices=sha256
CMSDigest=6611aefb3b6f5a6711a5f8bec81dc88a7ed8108cb86aa972d2b23320fed169d1
CMSDigestType=2
Executable Segment base=0
Executable Segment limit=49152
Executable Segment flags=0x1
Page size=4096
CDHash=6611aefb3b6f5a6711a5f8bec81dc88a7ed8108c
Signature=adhoc
Info.plist entries=9
TeamIdentifier=not set
Sealed Resources version=2 rules=13 files=1
Internal requirements count=0 size=12
zoziel:Downloads punisher$
```

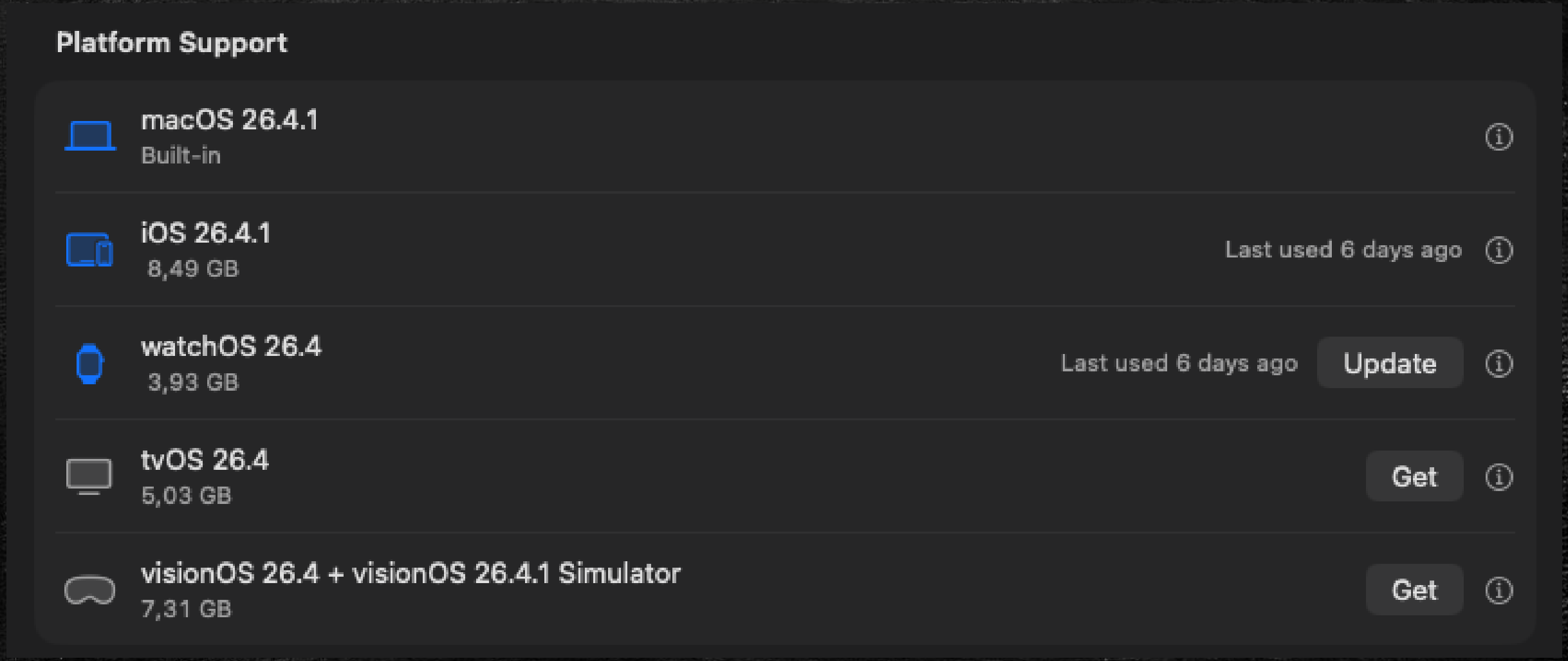
App - Assinado

```
zoziel:CleanOS punisher$ codesign -dv --verbose=4 CleanOSX2.app
Executable=/Users/punisher/Desktop/Apps/CleanOS/CleanOSX2.app/Contents/MacOS/CleanOS
Identifier=com.cleanos.app
Format=app bundle with Mach-O universal (x86_64 arm64)
CodeDirectory v=20500 size=2715 flags=0x10000(runtime) hashes=74+7 location=embedded
VersionPlatform=1
VersionMin=851968
VersionSDK=1704960
Hash type=sha256 size=32
CandidateCDHash sha256=33e5207bea4b3c50c5711219728ad33fd88f007a
CandidateCDHashFull sha256=33e5207bea4b3c50c5711219728ad33fd88f007a0ffa5f42a66ebe4648671f79
Hash choices=sha256
CMSDigest=33e5207bea4b3c50c5711219728ad33fd88f007a0ffa5f42a66ebe4648671f79
CMSDigestType=2
Executable Segment base=0
Executable Segment limit=1048576
Executable Segment flags=0x1
Page size=16384
CDHash=33e5207bea4b3c50c5711219728ad33fd88f007a
Signature size=9057
Authority=Developer ID Application: ZOZIEL PINTO FREIRE (7FG5338R44)
Authority=Developer ID Certification Authority
Authority=Apple Root CA
Timestamp=5 May 2026 at 22:07:52
Notarization Ticket=stapled
Info.plist entries=25
TeamIdentifier=7FG5338R44
Runtime Version=26.4.0
Sealed Resources version=2 rules=13 files=2
Internal requirements count=1 size=208
zoziel:CleanOS punisher$
```

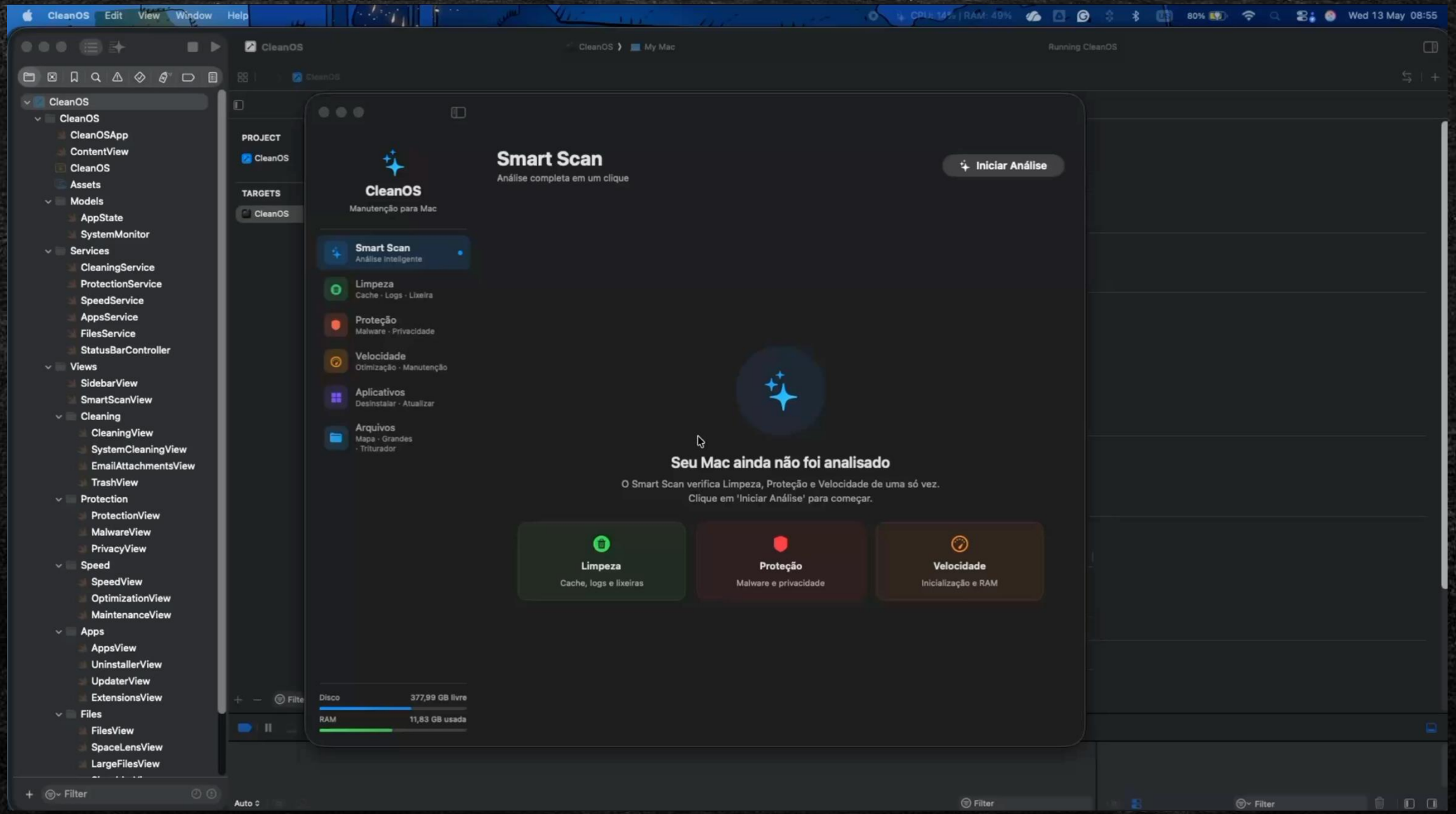


Simulação e Testes no Xcode

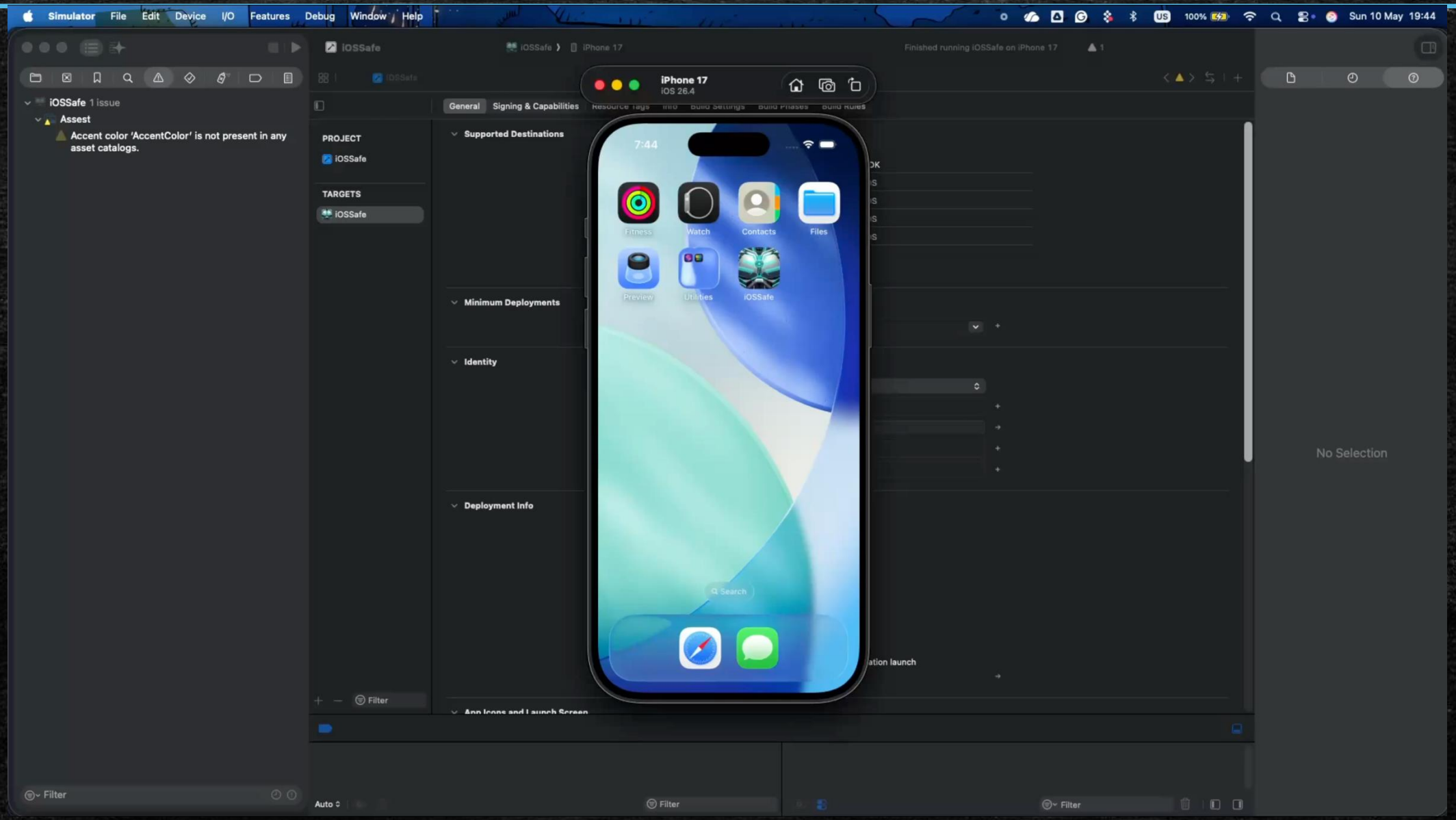
Xcode - Simuladores



Xcode – Simuladores- macOS



Xcode – Simuladores- iOS



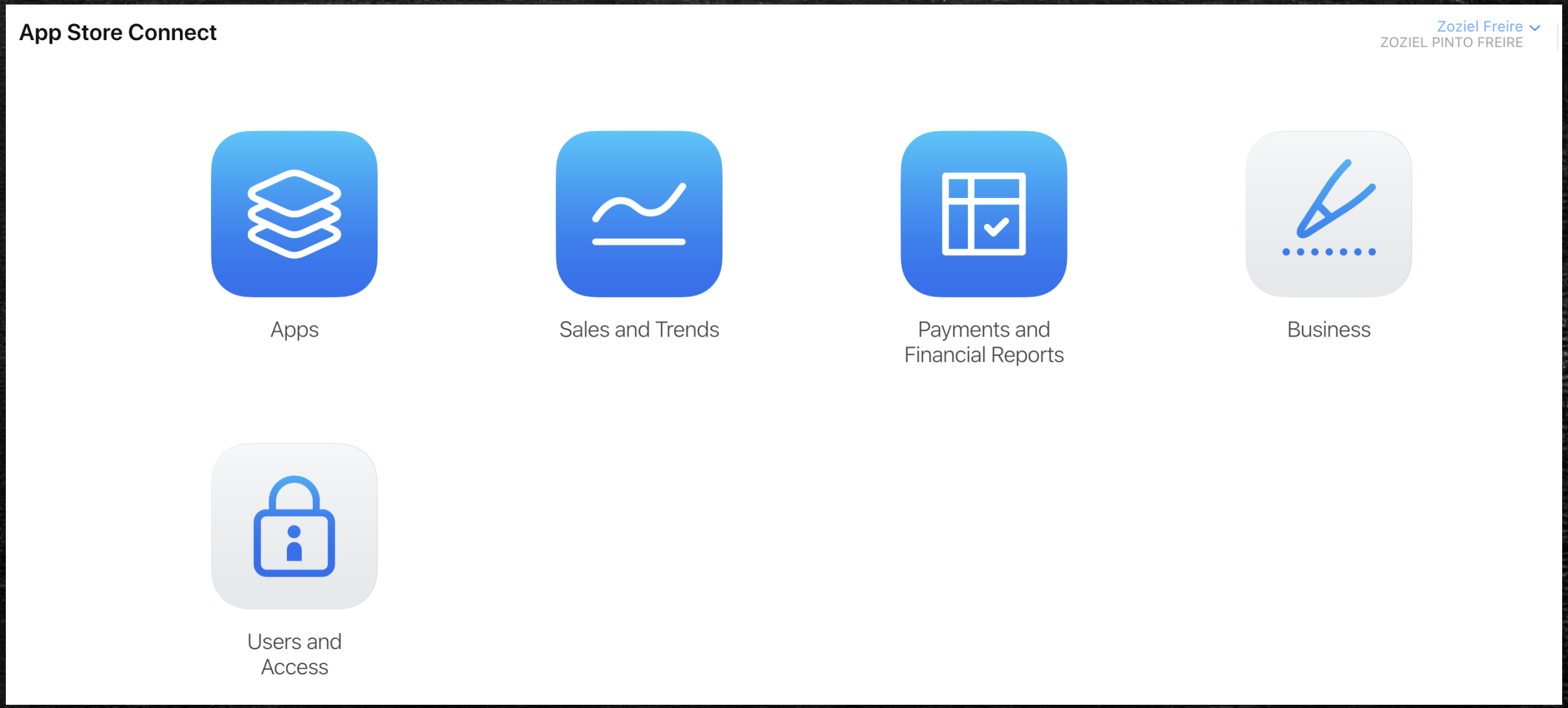
Xcode – Simuladores- iPadOS



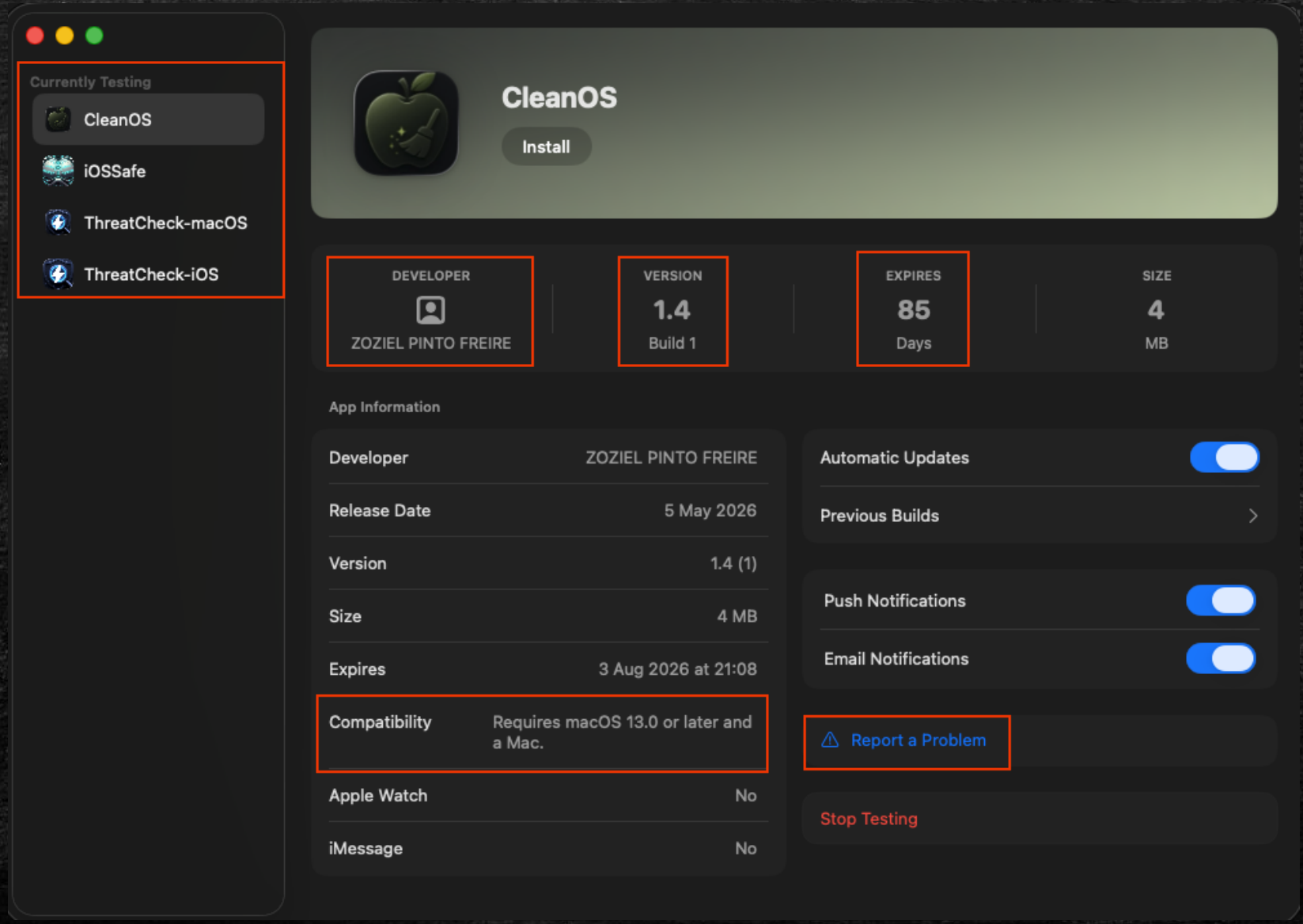
Xcode – Simuladores- WatchOS



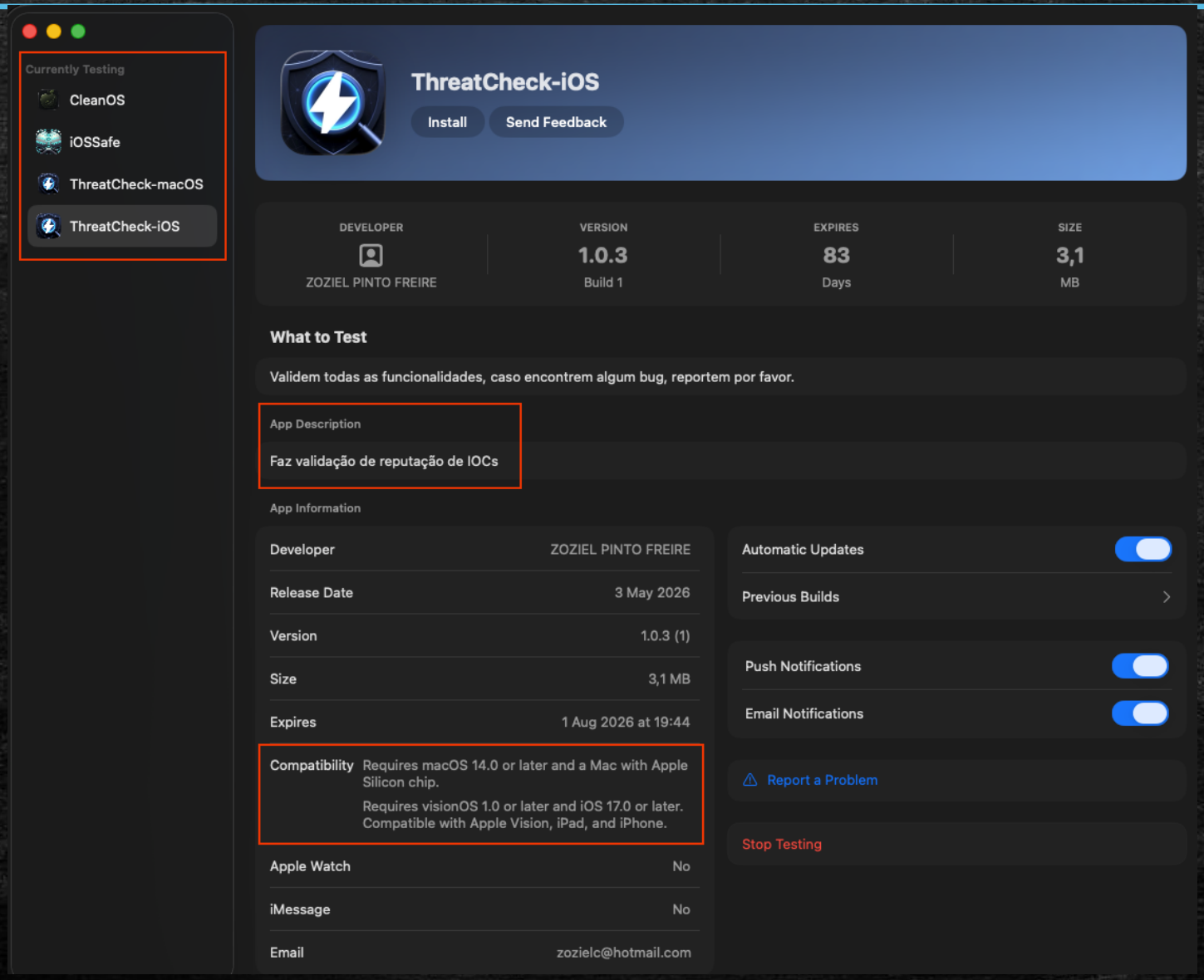
App Store Connect



TestFlight


Currently Testing
CleanOS
iOSSafe
ThreatCheck-macOS
ThreatCheck-iOS
CleanOS
Install
DEVELOPER
ZOZIEL PINTO FREIRE
VERSION
1.4
Build 1
EXPIRES
85
Days
SIZE
4
MB
App Information
Developer
ZOZIEL PINTO FREIRE
Release Date
5 May 2026
Version
1.4 (1)
Size
4 MB
Expires
3 Aug 2026 at 21:08
Compatibility
Requires macOS 13.0 or later and a Mac.
Apple Watch
No
iMessage
No
Automatic Updates
Previous Builds
Push Notifications
Email Notifications
Report a Problem
Stop Testing

TestFlight


Currently Testing
CleanOS
iOSSafe
ThreatCheck-macOS
ThreatCheck-iOS
ThreatCheck-iOS
Install
Send Feedback
DEVELOPER
ZOZIEL PINTO FREIRE
VERSION
1.0.3
Build 1
EXPIRES
83
Days
SIZE
3,1
MB
What to Test
Validem todas as funcionalidades, caso encontrem algum bug, reportem por favor.
App Description
Faz validação de reputação de IOCs
App Information
Developer
ZOZIEL PINTO FREIRE
Release Date
3 May 2026
Version
1.0.3 (1)
Size
3,1 MB
Expires
1 Aug 2026 at 19:44
Compatibility
Requires macOS 14.0 or later and a Mac with Apple Silicon chip.
Requires visionOS 1.0 or later and iOS 17.0 or later. Compatible with Apple Vision, iPad, and iPhone.
Apple Watch
No
iMessage
No
Email
zozielc@hotmail.com
Automatic Updates
Previous Builds
Push Notifications
Email Notifications
Report a Problem
Stop Testing



Conclusão

Conclusão

- Desenvolvimento Apple vai alem de escrever Código:
- Ecossistema Integrado
- Ferramentas maduras no Xcode
- Segurança e Privacidade como pilares
- Processo rigoroso de **Validação e Publicação**
- Experiência consistente para o **Usuário Final**

Ideia → Desenvolvimento → Testes → Validação → App Store

"Desenvolver para plataformas Apple é construir software com foco em qualidade, segurança e experiência"

Material

File System
https://developer.apple.com/library/archive/documentation/FileManagement/Conceptual/FileSystemProgrammingGuide/FileSystemOverview/FileSystemOverview.html

Security
https://developer.apple.com/documentation/security

Apple Platform Security Guide
https://help.apple.com/pdf/security/en_US/apple-platform-security-guide.pdf

Xcode Documentation
https://developer.apple.com/documentation/xcode

Swift Documentation
https://developer.apple.com/documentation/swift/

Apple Code Signing Overview
https://developer.apple.com/forums/thread/707080?utm_source=chatgpt.com

Material

LaunchAgents
https://support.apple.com/pt-br/guide/terminal/apdc6c1077b-5d5d-4d35-9c19-60f2397b2369/mac

The Art Of Infection In MacOS
https://hadess.io/the-art-of-infection-in-macos/

Books Patrick Wardle:

The Art of Mac Malware, Volume 1: The Guide to Analyzing Malicious Software
The Art of Mac Malware, Volume 2: Detecting Malicious Software

Blogs:
https://core-jmp.org/2026/03/edr-internals-for-macos-and-linux/
https://cocomelonc.github.io/categories/#macos
https://objective-see.org/blog.html
https://www.mac4n6.com/

Laboratório

Lab Tools

- Parallels
  - MacOS (15 Sequoia)
  - MacOS (26 Tahoe)
- Sublime
- Create-DMG
- Xcode
- Apple Developer Program
- App Store Connect
- Swift

Perguntas?

Contatos

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linkedin.com/in/zozielfreire

Obrigado!

https://discord.gg/tSpGtcUHVJ